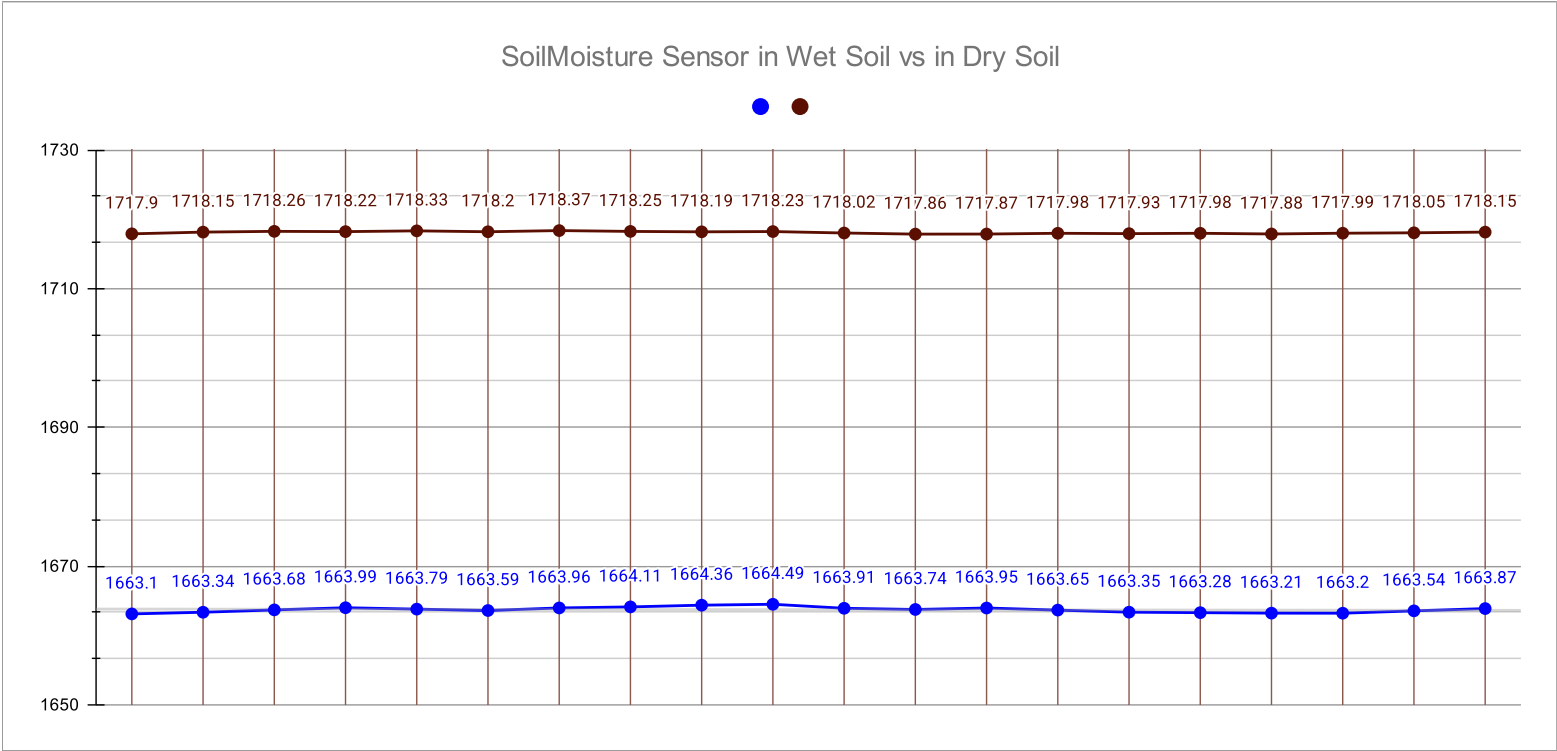


Testcase	AVG	StDev	MAX	MIN
SoilMoist_WetSoil	1663.7055	0.3868424812	1664.49	1663.1
SoilMoist_DrySoil	1718.0905	0.1628843632	1718.37	1717.86
SoilMoist_InWater	1689.8775	0.1324217743	1690.1	1689.71
SoilMoist_InAIR	1738.9865	0.1568195409	1739.26	1738.63

Constant Name	Value (in 12-bit ADC)	Mean in Soil Moisture
WET	1662	100%
DRY	1719	0%
-	-	-
-	-	-

Linear Interpolation
Soil Moisture = (ADC_raw - DRY) / (WET-DRY)

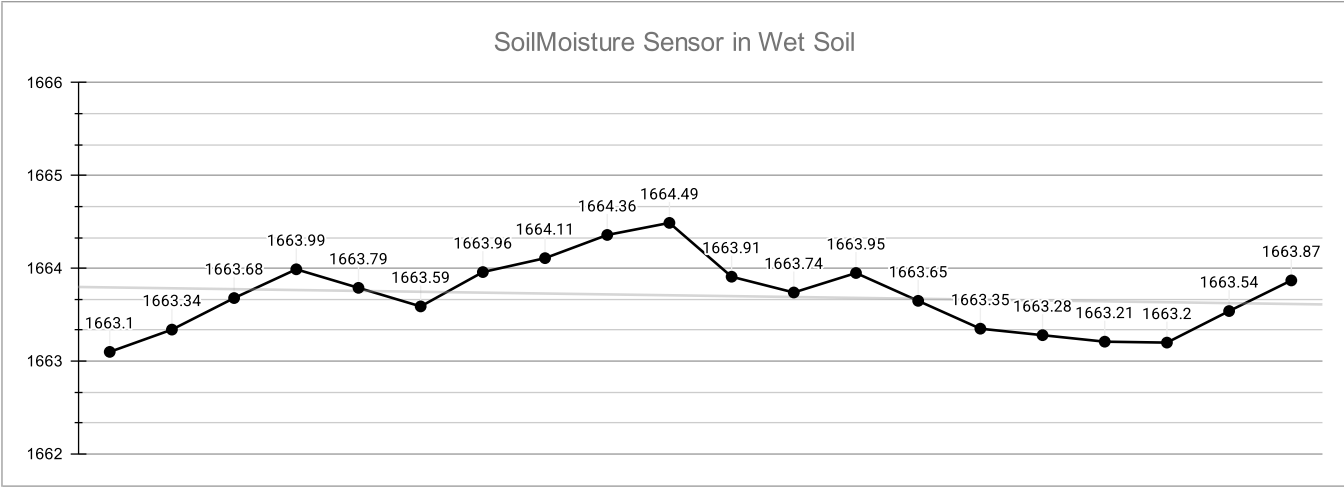
$$SoilMoisture(\%) = \frac{(ADC_{raw} - 1662)}{(1719 - 1662)} \cdot 100$$



WetSoil - DrySoil
-54.8
-54.81
-54.58
-54.23
-54.54
-54.61
-54.41
-54.14
-53.83
-53.74
-54.11
-54.12
-53.92
-54.33
-54.58
-54.7
-54.67
-54.79
-54.51
-54.28

AVG vs
-54.385

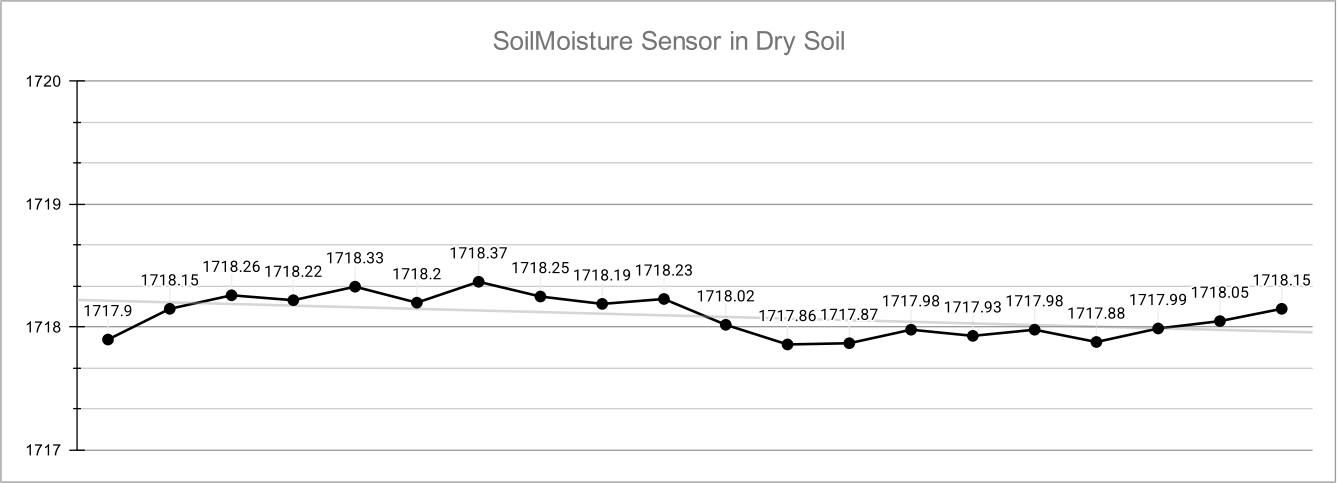
Timestamp	Sensor_Type	Value1	Value2
-	SoilMoist_WetSoil	1663.1	
-	SoilMoist_WetSoil	1663.34	
-	SoilMoist_WetSoil	1663.68	
-	SoilMoist_WetSoil	1663.99	
-	SoilMoist_WetSoil	1663.79	
-	SoilMoist_WetSoil	1663.59	
-	SoilMoist_WetSoil	1663.96	
-	SoilMoist_WetSoil	1664.11	
-	SoilMoist_WetSoil	1664.36	
-	SoilMoist_WetSoil	1664.49	
-	SoilMoist_WetSoil	1663.91	
-	SoilMoist_WetSoil	1663.74	
-	SoilMoist_WetSoil	1663.95	
-	SoilMoist_WetSoil	1663.65	
-	SoilMoist_WetSoil	1663.35	
-	SoilMoist_WetSoil	1663.28	
-	SoilMoist_WetSoil	1663.21	
-	SoilMoist_WetSoil	1663.2	
-	SoilMoist_WetSoil	1663.54	
-	SoilMoist_WetSoil	1663.87	



log 22:14:18.401 -> => SoilMoist_WetSoil[0]: 1663.099976
22:14:19.008 -> => SoilMoist_WetSoil[1]: 1663.344971
22:14:19.612 -> => SoilMoist_WetSoil[2]: 1663.677734
22:14:20.207 -> => SoilMoist_WetSoil[3]: 1663.993896
22:14:20.816 -> => SoilMoist_WetSoil[4]: 1663.794238
22:14:21.410 -> => SoilMoist_WetSoil[5]: 1663.589478
22:14:22.001 -> => SoilMoist_WetSoil[6]: 1663.959961
22:14:22.622 -> => SoilMoist_WetSoil[7]: 1664.111938
22:14:23.179 -> => SoilMoist_WetSoil[8]: 1664.356323
22:14:23.780 -> => SoilMoist_WetSoil[9]: 1664.488525
22:14:24.378 -> => SoilMoist_WetSoil[10]: 1663.914038
22:14:25.002 -> => SoilMoist_WetSoil[11]: 1663.738281
22:14:25.611 -> => SoilMoist_WetSoil[12]: 1663.951416
22:14:26.215 -> => SoilMoist_WetSoil[13]: 1663.653809
22:14:26.807 -> => SoilMoist_WetSoil[14]: 1663.346069
22:14:27.414 -> => SoilMoist_WetSoil[15]: 1663.278809
22:14:28.005 -> => SoilMoist_WetSoil[16]: 1663.214844
22:14:28.581 -> => SoilMoist_WetSoil[17]: 1663.204102
22:14:29.178 -> => SoilMoist_WetSoil[18]: 1663.543945
22:14:29.812 -> => SoilMoist_WetSoil[19]: 1663.866699

AVG	1663.7055
StDev	0.3868424812
MAX	1664.49
MIN	1663.1

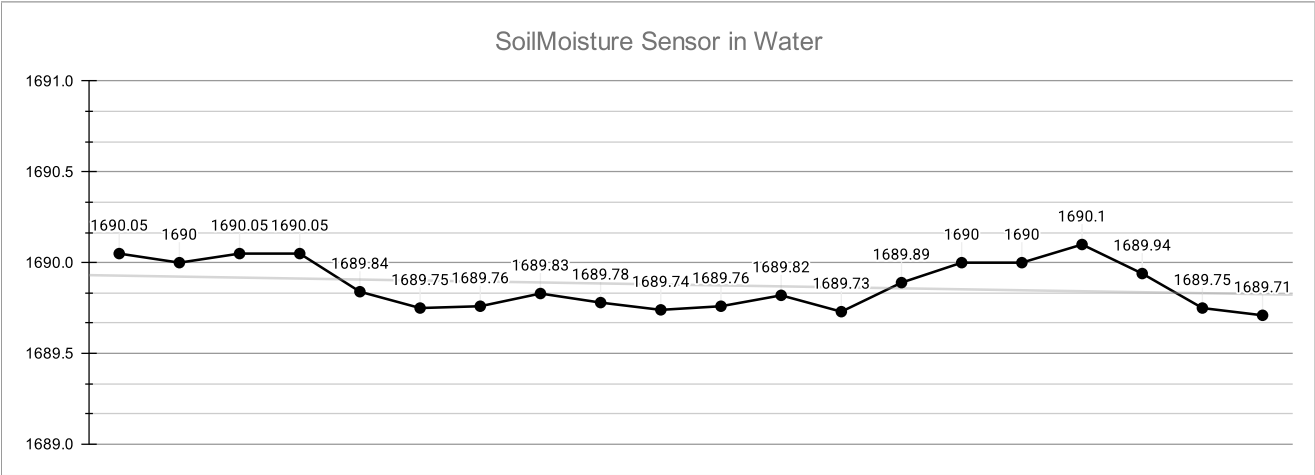
Timestamp	Sensor_Type	Value1	Value2
-	SoilMoist_DrySoil	1717.9	
-	SoilMoist_DrySoil	1718.15	
-	SoilMoist_DrySoil	1718.26	
-	SoilMoist_DrySoil	1718.22	
-	SoilMoist_DrySoil	1718.33	
-	SoilMoist_DrySoil	1718.2	
-	SoilMoist_DrySoil	1718.37	
-	SoilMoist_DrySoil	1718.25	
-	SoilMoist_DrySoil	1718.19	
-	SoilMoist_DrySoil	1718.23	
-	SoilMoist_DrySoil	1718.02	
-	SoilMoist_DrySoil	1717.86	
-	SoilMoist_DrySoil	1717.87	
-	SoilMoist_DrySoil	1717.98	
-	SoilMoist_DrySoil	1717.93	
-	SoilMoist_DrySoil	1717.98	
-	SoilMoist_DrySoil	1717.88	
-	SoilMoist_DrySoil	1717.99	
-	SoilMoist_DrySoil	1718.05	
-	SoilMoist_DrySoil	1718.15	



log 22:03:07.403 -> => SoilMoist_DrySoil[0]: 1717.800466
22:03:08.003 -> => SoilMoist_DrySoil[1]: 1718.149976
22:03:08.626 -> => SoilMoist_DrySoil[2]: 1718.257446
22:03:09.232 -> => SoilMoist_DrySoil[3]: 1718.219580
22:03:09.840 -> => SoilMoist_DrySoil[4]: 1718.328540
22:03:10.448 -> => SoilMoist_DrySoil[5]: 1718.197168
22:03:11.025 -> => SoilMoist_DrySoil[6]: 1718.367310
22:03:11.615 -> => SoilMoist_DrySoil[7]: 1718.248901
22:03:12.203 -> => SoilMoist_DrySoil[8]: 1718.186401
22:03:12.830 -> => SoilMoist_DrySoil[9]: 1718.227051
22:03:13.420 -> => SoilMoist_DrySoil[10]: 1718.015747
22:03:14.024 -> => SoilMoist_DrySoil[11]: 1717.864990
22:03:14.603 -> => SoilMoist_DrySoil[12]: 1717.871704
22:03:15.234 -> => SoilMoist_DrySoil[13]: 1717.978149
22:03:15.803 -> => SoilMoist_DrySoil[14]: 1717.929199
22:03:16.434 -> => SoilMoist_DrySoil[15]: 1717.982788
22:03:17.039 -> => SoilMoist_DrySoil[16]: 1717.883667
22:03:17.642 -> => SoilMoist_DrySoil[17]: 1717.989429
22:03:18.215 -> => SoilMoist_DrySoil[18]: 1718.049951
22:03:18.803 -> => SoilMoist_DrySoil[19]: 1718.152515

AVG	1718.0905
StDev	0.1628843632
MAX	1718.37
MIN	1717.86

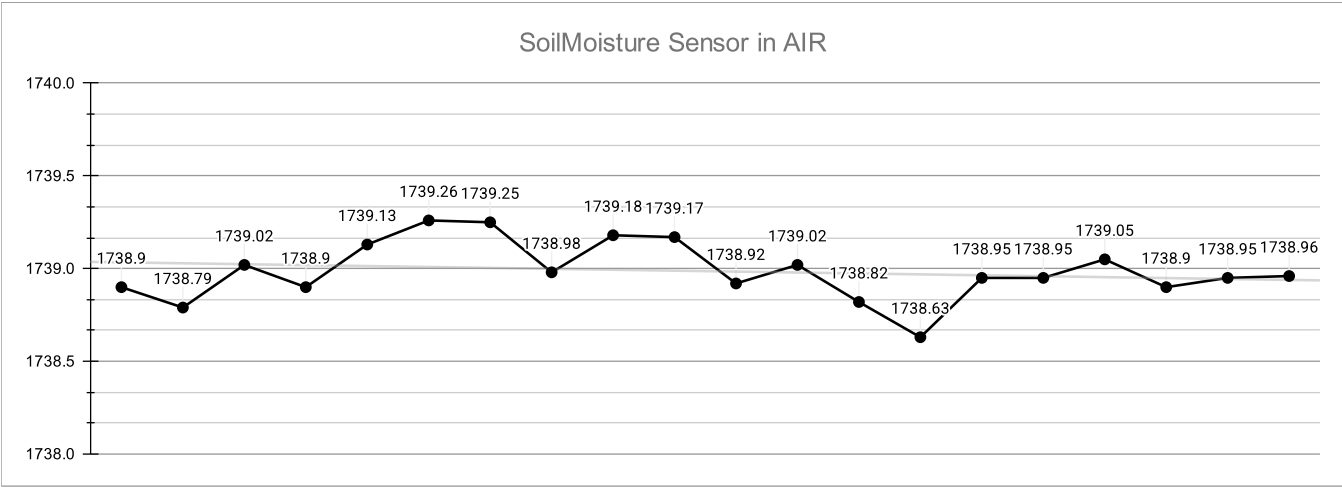
Timestamp	Sensor_Type	Value1	Value2
-	SoilMoist_InWater	1690.05	
-	SoilMoist_InWater	1690	
-	SoilMoist_InWater	1690.05	
-	SoilMoist_InWater	1690.05	
-	SoilMoist_InWater	1689.84	
-	SoilMoist_InWater	1689.75	
-	SoilMoist_InWater	1689.76	
-	SoilMoist_InWater	1689.83	
-	SoilMoist_InWater	1689.78	
-	SoilMoist_InWater	1689.74	
-	SoilMoist_InWater	1689.76	
-	SoilMoist_InWater	1689.82	
-	SoilMoist_InWater	1689.73	
-	SoilMoist_InWater	1689.89	
-	SoilMoist_InWater	1690	
-	SoilMoist_InWater	1690	
-	SoilMoist_InWater	1690.1	
-	SoilMoist_InWater	1689.94	
-	SoilMoist_InWater	1689.75	
-	SoilMoist_InWater	1689.71	



log	21:46:04.447 -> ==> SoilMoist_InWater[0]: 1690.050049
	21:46:05.055 -> ==> SoilMoist_InWater[1]: 1689.997559
	21:46:05.645 -> ==> SoilMoist_InWater[2]: 1690.047729
	21:46:06.249 -> ==> SoilMoist_InWater[3]: 1690.045288
	21:46:06.857 -> ==> SoilMoist_InWater[4]: 1689.843018
	21:46:07.466 -> ==> SoilMoist_InWater[5]: 1689.750854
	21:46:08.055 -> ==> SoilMoist_InWater[6]: 1689.763306
	21:46:08.657 -> ==> SoilMoist_InWater[7]: 1689.825195
	21:46:09.248 -> ==> SoilMoist_InWater[8]: 1689.783936
	21:46:09.855 -> ==> SoilMoist_InWater[9]: 1689.744751
	21:46:10.458 -> ==> SoilMoist_InWater[10]: 1689.757568
	21:46:11.067 -> ==> SoilMoist_InWater[11]: 1689.819702
	21:46:11.620 -> ==> SoilMoist_InWater[12]: 1689.728760
	21:46:12.245 -> ==> SoilMoist_InWater[13]: 1689.892334
	21:46:12.822 -> ==> SoilMoist_InWater[14]: 1689.997681
	21:46:13.420 -> ==> SoilMoist_InWater[15]: 1689.997803
	21:46:14.046 -> ==> SoilMoist_InWater[16]: 1690.097900
	21:46:14.648 -> ==> SoilMoist_InWater[17]: 1689.942993
	21:46:15.254 -> ==> SoilMoist_InWater[18]: 1689.745850
	21:46:15.819 -> ==> SoilMoist_InWater[19]: 1689.708618

AVG	1689.8775
StDev	0.1324217743
MAX	1690.1
MIN	1689.71

Timestamp	Sensor_Type	Value1	Value2
-	SoilMoist_InAIR	1738.9	
-	SoilMoist_InAIR	1738.79	
-	SoilMoist_InAIR	1739.02	
-	SoilMoist_InAIR	1738.9	
-	SoilMoist_InAIR	1739.13	
-	SoilMoist_InAIR	1739.26	
-	SoilMoist_InAIR	1739.25	
-	SoilMoist_InAIR	1738.98	
-	SoilMoist_InAIR	1739.18	
-	SoilMoist_InAIR	1739.17	
-	SoilMoist_InAIR	1738.92	
-	SoilMoist_InAIR	1739.02	
-	SoilMoist_InAIR	1738.82	
-	SoilMoist_InAIR	1738.63	
-	SoilMoist_InAIR	1738.95	
-	SoilMoist_InAIR	1738.95	
-	SoilMoist_InAIR	1739.05	
-	SoilMoist_InAIR	1738.9	
-	SoilMoist_InAIR	1738.95	
-	SoilMoist_InAIR	1738.96	



log

21:35:07.035 -> => SoilMoist_InAIR[0]: 1738.900049
21:35:07.640 -> => SoilMoist_InAIR[1]: 1738.785010
21:35:08.197 -> => SoilMoist_InAIR[2]: 1739.015796
21:35:08.839 -> => SoilMoist_InAIR[3]: 1739.900024
21:35:09.397 -> => SoilMoist_InAIR[4]: 1739.130078
21:35:10.020 -> => SoilMoist_InAIR[5]: 1739.263525
21:35:10.627 -> => SoilMoist_InAIR[6]: 1739.245361
21:35:11.196 -> => SoilMoist_InAIR[7]: 1738.983154
21:35:11.796 -> => SoilMoist_InAIR[8]: 1739.183960
21:35:12.419 -> => SoilMoist_InAIR[9]: 1739.174805
21:35:13.025 -> => SoilMoist_InAIR[10]: 1738.916016
21:35:13.613 -> => SoilMoist_InAIR[11]: 1739.020264
21:35:14.218 -> => SoilMoist_InAIR[12]: 1738.819214
21:35:14.795 -> => SoilMoist_InAIR[13]: 1738.628296
21:35:15.432 -> => SoilMoist_InAIR[14]: 1738.946899
21:35:15.995 -> => SoilMoist_InAIR[15]: 1738.949585
21:35:16.616 -> => SoilMoist_InAIR[16]: 1739.052124
21:35:17.224 -> => SoilMoist_InAIR[17]: 1738.899536
21:35:17.816 -> => SoilMoist_InAIR[18]: 1738.954590
21:35:18.395 -> => SoilMoist_InAIR[19]: 1738.956909

AVG	1738.9865
StDev	0.1568195409
MAX	1739.26
MIN	1738.63